

Several teens visited the latest Bizaarre Bazaar, where representatives of movies, series, and performers reveal new items for their followers. Each person took advantage of the low prices to pick up new ways to support their favorites.

All shirts were \$13, hats were \$9, and stickers were \$2. Figure out how many of each item a student bought, and their total cost.



Input

The first line is a student name (one word), followed by an integer N: the number of vendors they visited. The next N lines each have 3 integers, for the count of shirts, hats, and stickers that student bought at each vendor.

```
Tabitha 5
1 4 8
0 1 8
0 0 4
0 5 0
0 3 0
```

Output

Find the total of each item and total cost for the student. Print one line as below. Make an item name plural if its count is not 1 ("shirt" below is singular, while others are plural.)

```
Tabitha spent $170 on 1 shirt, 13 hats, and 20 stickers.
```

Discussion

All integers in the input will be from 0 to 20. Do not use commas or decimal points in output.

Additional Example

Input 2	Output 2
Mauricio 4 19 0 0 20 0 0 18 0 0 20 1 0	Mauricio spent \$1010 on 77 shirts, 1 hat, and 0 stickers